

List A3

$$1- A = (a_{ij})_{3 \times 3} \rightarrow a_{ij} = \begin{cases} i+j, & i < j \\ 2i-i, & i = j \\ j-i, & i > j \end{cases}$$

$$a_{ij} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$A = \begin{pmatrix} 2 \cdot 1 - 1 & 1 + 2 & 1 + 3 \\ 1 - 2 & 2 \cdot 2 - 2 & 2 + 3 \\ 1 - 3 & 2 - 3 & 2 \cdot 3 - 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 4 \\ -1 & 2 & 5 \\ -2 & -1 & 3 \end{pmatrix}$$

$$AX = B \Leftrightarrow X = BA^{-1}$$

$$\text{Det } A = \begin{vmatrix} 1 & 3 & 4 \\ -1 & 2 & 5 \\ -2 & -1 & 3 \end{vmatrix}$$

$$\text{inverse} = \frac{1}{\det} \cdot \text{adj}$$

$$+16 + 5 + 9 + 6 + 4 - 30 = 10$$

matriz adjunta:

$$a_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 5 \\ -1 & 3 \end{vmatrix} = +11 \quad a_{21} = (-1)^{2+1} = \begin{vmatrix} 3 & 4 \\ -1 & 3 \end{vmatrix} = -13$$

$$a_{12} = (-1)^{1+2} \cdot \begin{vmatrix} -1 & 5 \\ -2 & 3 \end{vmatrix} = -7 \quad a_{22} = (-1)^{2+2} = \begin{vmatrix} 1 & 4 \\ -2 & 3 \end{vmatrix} = +11$$

$$a_{13} = (-1)^{1+3} \cdot \begin{vmatrix} -1 & 2 \\ -2 & -1 \end{vmatrix} = +5 \quad a_{23} = (-1)^{2+3} = \begin{vmatrix} 1 & 3 \\ -2 & -1 \end{vmatrix} = -5$$

$$a_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 3 & 4 \\ 2 & 5 \end{vmatrix} = +7 \quad a_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 4 \\ -1 & 5 \end{vmatrix} = 9 \quad a_{33} = (-1)^{3+3} = \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} = +5$$

$$\text{Ca} = \begin{pmatrix} +11 & -7 & +5 \\ -13 & +11 & -5 \\ +7 & -9 & +5 \end{pmatrix} \Rightarrow \text{adj} = \begin{pmatrix} +11 & -13 & +7 \\ -7 & +11 & -9 \\ +5 & -5 & +5 \end{pmatrix} \begin{matrix} 1 \\ \det \\ \hookrightarrow 10 \end{matrix}$$

$$A^{-1} = \begin{pmatrix} +1,1 & -1,3 & +0,7 \\ -0,7 & +1,1 & -0,9 \\ +0,5 & -0,5 & +0,5 \end{pmatrix}$$



$$X = \begin{pmatrix} +1,1 & -1,3 & +0,7 \\ -0,7 & +1,1 & -0,9 \\ +0,5 & -0,5 & +0,5 \end{pmatrix}_{3 \times 3} \times \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}_{3 \times 1}$$

$$\begin{pmatrix} 1,1 \cdot 1 + (-1,3) \cdot 2 + 0,7 \cdot (-3) \\ -0,7 \cdot 1 + 1,1 \cdot 2 + (-0,9) \cdot (-3) \\ 0,5 \cdot 1 + (-0,5) \cdot 2 + 0,5 \cdot (-3) \end{pmatrix} = \begin{pmatrix} 1,1 - 2,6 - 2,1 \\ -0,7 + 2,2 + 2,7 \\ 0,5 - 1 - 1,5 \end{pmatrix}$$

$$X = \begin{pmatrix} -3,6 \\ 4,2 \\ -2 \end{pmatrix}$$

$$2-a) \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} X = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$AX = B$$

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B$$

$$\therefore X = A^{-1}B$$

$$\text{Det } |A| = -5$$

$$X = \begin{pmatrix} -\frac{3}{5} & +\frac{4}{5} \\ +\frac{2}{5} & -\frac{1}{5} \end{pmatrix}_{2 \times 2} \times \begin{pmatrix} 2 \\ -1 \end{pmatrix}_{2 \times 1}$$

$$X = \begin{pmatrix} -\frac{3}{5} \cdot 2 + \frac{4}{5} \cdot (-1) \\ \frac{2}{5} \cdot 2 + (-\frac{1}{5}) \cdot (-1) \end{pmatrix} = X = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$5) \begin{pmatrix} 2 & 3 \\ 5 & 5 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} Y = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$$

$$AX = B \rightarrow X = A^{-1}B$$

• inversa:

$$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} Y = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 3 \\ 5 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \rightarrow \text{Det} = -1$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} Y = \begin{pmatrix} -1 & 0 \\ -3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} \frac{5}{-1} & -\frac{2}{-1} \\ -\frac{3}{-1} & \frac{1}{-1} \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$Y = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 \\ -3 & 2 \end{pmatrix}$$

$$Y = \begin{pmatrix} -5 \cdot (-1) + 2 \cdot (-3) & -5 \cdot 0 + 2 \cdot 2 \\ 3 \cdot (-1) + (-1) \cdot (-3) & 3 \cdot 0 + (-1) \cdot 2 \end{pmatrix} = Y = \begin{pmatrix} -1 & 4 \\ 0 & -2 \end{pmatrix}$$

$$c) \begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 2 & 3 & 1 \end{pmatrix} W = \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 2 & 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix}$$

$$I - 1 \cdot a = 5$$

$$II - 2 \cdot a - b = 7$$

$$2 \cdot 5 - b = 7$$

$$10 - b = 7$$

$$b = 3$$

$$III - 2a + 3b + c = 2 \quad \therefore W = \begin{pmatrix} 5 \\ 3 \\ -17 \end{pmatrix}$$

$$2 \cdot 5 + 3 \cdot 3 + c = 2$$

$$10 + 9 + c = 2$$

$$c = -17$$

$$3- a) A \times B = C$$

$$A^{-1} A \times B = A^{-1} C$$

$$I \times B B^{-1} = A^{-1} C B^{-1}$$

$$X = A^{-1} C B^{-1}$$

$$b) A(B+X) = A$$

$$A^{-1} A(B+X) = A^{-1} A$$

$$I(B+X) = I$$

$$B+X = I$$

$$X = I - B$$

$$c) A C \times B = C$$

$$A^{-1} A C \times B = A^{-1} C$$

$$I C C^{-1} \times B = A^{-1} C C^{-1} C$$

$$I \times B = A^{-1} I$$

$$X B B^{-1} = A^{-1} I B^{-1}$$

$$X = A^{-1} B^{-1}$$

$$d) (AB)^{-1} (AX) = C C^{-1}$$

$$(AB) (AB)^{-1} (AX) = (AB) I$$

$$I (AX) = AB$$

$$AX = AB$$

$$A^{-1} A X = A^{-1} A B$$

$$X = B$$

$$e) A B^T \times B^{-1} = A^T$$

$$A A^{-1} B^T \times B^{-1} = A^{-1} A^T$$

$$I B^T B^{-1} \times B^{-1} = A^{-1} B^{-1} A^T$$

$$I \times B^{-1} B = A^{-1} B^{-1} A^T B$$

$$I \times I = A^{-1} B^{-1} A^T B$$

$$X = A^{-1} B^{-1} A^T B$$

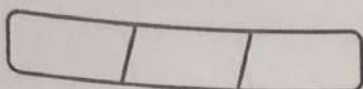
$$f) 2AX - X = 3B$$

$$(2A - I) X = 3B$$

$$(2A - I)^{-1} (2A - I) X = (2A - I)^{-1} 3B$$

$$I X = (2A - I)^{-1} 3B$$

$$X = (2A - I)^{-1} 3B$$



$$4-a) \begin{cases} 3x - 4y = 1 \\ 2x + 6y = 18 \end{cases} \quad \begin{matrix} A & X & B \\ \begin{pmatrix} 3 & -4 \\ 2 & 6 \end{pmatrix} & \begin{pmatrix} x \\ y \end{pmatrix} & = \begin{pmatrix} 1 \\ 18 \end{pmatrix} \end{matrix}$$

$$\det A = |A| = 26$$

$$|Ax| = \begin{vmatrix} 1 & -4 \\ 18 & 6 \end{vmatrix} = 78 \quad x = \frac{\det x}{\det A} = \frac{78}{26} = 3$$

$$|Ay| = \begin{vmatrix} 3 & 1 \\ 2 & 18 \end{vmatrix} = 52 \quad y = \frac{\det y}{\det A} = \frac{52}{26} = 2$$

$$b) \begin{cases} 5x + 8y = 34 \\ 10x + 16y = 50 \end{cases} \quad \begin{pmatrix} 5 & 8 \\ 10 & 16 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 34 \\ 50 \end{pmatrix}$$

$$\det |A| = 80 - 80 = 0 \rightarrow \text{Sistema Impossível (S.I.)}$$

$$c) \begin{cases} x + 2y = 5 \\ 2x - 3y = -9 \end{cases} \quad \begin{matrix} A & X & D \\ \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix} & \begin{pmatrix} x \\ y \end{pmatrix} & = \begin{pmatrix} 5 \\ -9 \end{pmatrix} \end{matrix}$$

$$\det = -3 - 4 = -7$$

$$|Ax| = \begin{vmatrix} 5 & 2 \\ -9 & -3 \end{vmatrix} = -15 + 18 = -3 \quad x = \frac{\det x}{\det A} = \frac{-3}{-7} = \frac{3}{7}$$

$$|Ay| = \begin{vmatrix} 1 & 5 \\ 2 & -9 \end{vmatrix} = -9 - 10 = -19 \quad y = \frac{\det y}{\det A} = \frac{-19}{-7} = \frac{19}{7}$$

$$d) \begin{cases} 3x + 2y - 5z = 8 \\ 2x - 4y - 2z = -4 \\ x - 2y - 3z = -9 \end{cases} \quad \begin{pmatrix} 3 & 2 & -5 \\ 2 & -4 & -2 \\ 1 & -2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \\ -9 \end{pmatrix}$$

$$\det(A) = 36 + 20 - 4 - 20 - 12 + 12$$

$$\det(A) = 32$$



$$\text{Det}(A) = 32$$

$$\text{Det } x = \begin{vmatrix} 8 & 2 & -5 \\ -4 & -4 & -2 \\ -4 & -2 & -3 \end{vmatrix} = \begin{vmatrix} 2 & 1 & -5 \\ -1 & -2 & -2 \\ -1 & -1 & -3 \end{vmatrix} \quad 4 \cdot 2 \Rightarrow 12 + 2 - 5 + 10 - 4 - 3$$

$$8 \cdot 12 = 96$$

$$\therefore x = \frac{96}{32} = 3$$

$$\text{Det } y = \begin{vmatrix} 3 & 8 & -5 \\ 2 & -4 & -2 \\ 1 & -4 & -3 \end{vmatrix} = \begin{vmatrix} 3 & 9 & -5 \\ 1 & -1 & -1 \\ 1 & -2 & -3 \end{vmatrix} \quad 4 \cdot 2 = 9 + 10 - 4 - 5 = 6 + 12$$

$$64$$

$$\therefore y = \frac{64}{32} = 2$$

$$\text{Det } z = \begin{vmatrix} 3 & 2 & 8 \\ 2 & -4 & -4 \\ 1 & -2 & -4 \end{vmatrix} = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -2 & -1 \\ 1 & -1 & -1 \end{vmatrix} \quad 4 \cdot 2 = 6 - 9 - 1 + 4 - 3 + 2$$

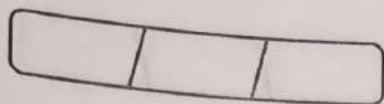
$$8 \cdot 4 = 32$$

$$\therefore z = \frac{32}{32} = 1$$

$$e) \begin{cases} x + 2y - z = 2 \\ 2x - y + 3z = 9 \\ 3x + 3y - 2z = 3 \end{cases} \rightarrow \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ 3 & 3 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 9 \\ 3 \end{pmatrix}$$

$$\text{Det}(A) = \begin{vmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ 3 & 3 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \\ 3 & 3 \end{vmatrix} = 2 + 18 - 6 - 3 - 9 + 8 = 10$$

$$\text{Det}(x) = \begin{vmatrix} 2 & 2 & -1 \\ 9 & -1 & 3 \\ 3 & 3 & -2 \end{vmatrix} = 4 - 12 + 18 - 3 - 18 + 16 = 35 = 3,5$$



$$\text{Det } y = \begin{vmatrix} 1 & 2 & -1 \\ 2 & 9 & 3 \\ 3 & 3 & -2 \end{vmatrix} = -18 - 6 + 18 + 27 - 9 + 8 = 20 = \frac{20}{10} = 2$$

$$\text{Det } z = \begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 9 \\ 3 & 3 & 3 \end{vmatrix} = -3 + 12 + 54 + 6 - 27 - 12 = 30 = \frac{30}{10} = 3$$

$$f) \begin{cases} x + 3z + 6 = -2 \\ 2x - 4y = -4 \\ 3x - 2y - 5z = 26 \end{cases} \rightarrow \begin{cases} x + 0y + 3z = -8 \\ 2x - 4y + 0z = -4 \\ 3x - 2y - 5z = 26 \end{cases}$$

$$\begin{pmatrix} 1 & 0 & 3 \\ 2 & -4 & 0 \\ 3 & -2 & -5 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -8 \\ -4 \\ 26 \end{pmatrix} \rightarrow \begin{vmatrix} 1 & 0 & 3 \\ 2 & -4 & 0 \\ 3 & -2 & -5 \end{vmatrix} \cdot \text{Det } |A| = \begin{vmatrix} 1 & 0 & 3 \\ 2 & -4 & 0 \\ 3 & -2 & -5 \end{vmatrix} = 20 + 0 - 12 + 36 = 44$$

$$\text{Det } |x| = \begin{vmatrix} -8 & 0 & 3 \\ -4 & -4 & 0 \\ 26 & -2 & -5 \end{vmatrix} = 2 \cdot 2 = 4 \quad \begin{vmatrix} -4 & 0 & 3 \\ -1 & -2 & 0 \\ 13 & -2 & -5 \end{vmatrix} = -40 + 6 + 0 + 79 = 44$$

$$\therefore x = \frac{\text{Det } x}{\text{Det } A} = \frac{4}{44} = 4$$

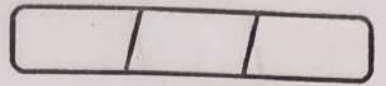
$$\text{Det } y = \begin{vmatrix} 1 & -8 & 3 \\ 2 & -4 & 0 \\ 3 & 26 & -5 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 3 \\ 1 & -1 & 0 \\ 3 & 13 & -5 \end{vmatrix} = 5 + 39 + 0 + 9 - 20 = 33 - 9 = 132$$

$$\therefore y = \frac{\text{Det } y}{\text{Det } A} = \frac{132}{44} = 3$$

$$\therefore z = \frac{176}{44} = 4$$

$$\text{Det } z = \begin{vmatrix} 1 & 0 & -8 \\ 2 & -4 & -4 \\ 3 & -2 & 26 \end{vmatrix} = 8 \cdot \begin{vmatrix} 1 & 0 & -4 \\ 1 & -1 & -1 \\ 3 & -1 & 13 \end{vmatrix} = -13 + 4 + 0 - 12 - 1 + 0 = 22 - 8 = 176$$





5-a)
$$\begin{cases} 3x_1 - 9x_2 = 0 \\ -6x_1 + 8x_2 = 0 \end{cases} \rightarrow \begin{pmatrix} 3 & -9 \\ -6 & 8 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{Det } A = \begin{vmatrix} 3 & -9 \\ -6 & 8 \end{vmatrix} = 3 \cdot 8 - (-9) \cdot (-6) = 24 - 24 = 0$$

$$\text{Det } x_1 = \begin{vmatrix} 0 & -9 \\ 0 & 8 \end{vmatrix} = 0 \quad \therefore \text{O Sistema é SPI}$$

Sistema possível Indeterminado

b)
$$\begin{cases} x + y + z = 0 \\ 2x + 2y + 4z = 0 \\ x + y + 3z = 0 \end{cases} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \\ 1 & 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Det } A = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \\ 1 & 1 & 3 \end{vmatrix} = 6 + 2 + 4 - 2 - 9 - 6 = 0$$

Det A = 0

$$\text{Det } x = \begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 3 \end{vmatrix} = 0 \quad \text{Det } y = 0 \text{ e } \text{Det } z = 0$$

$\therefore$ O sistema é possível e Indeterminado
SPI

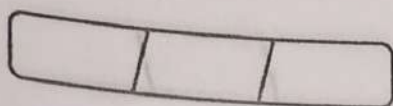
c)
$$\begin{cases} x + y + 2z = 0 \\ x - y - 3z = 0 \\ x + 4y = 0 \end{cases} \rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & -3 \\ 1 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Det } A = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & -3 \\ 1 & 4 & 0 \end{vmatrix} = 0 + 8 - 3 + 2 + 12 + 0 = 19$$

$\Rightarrow \text{Det } A = 19$

$\therefore$ O sistema é possível e Determinado
SPD





$$6-a) \begin{cases} 3x + my = 2 \\ x - y = 1 \end{cases} \rightarrow \begin{pmatrix} 3 & m \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

para SPD $\rightarrow \text{Det} A \neq 0$

$$\text{Det } A = \begin{vmatrix} 3 & m \\ 1 & 1 \end{vmatrix} = 3 - m \neq 0 \quad \therefore S = \{m \in \mathbb{R} / m \neq -3\}$$
$$m \neq -3$$

$$b) \begin{cases} 3x + 2(m-1)y = 1 \\ mx - 9y = 0 \end{cases} \Rightarrow \begin{cases} 3x + (2m-2)y = 1 \\ mx - 9y = 0 \end{cases} \Rightarrow \begin{cases} 3x + 2my - 2y = 1 \\ mx - 9y = 0 \end{cases}$$

$$\begin{pmatrix} 3 & 2m-2 \\ m & -9 \end{pmatrix} \neq 0 \Rightarrow -12 - (m(2m-2)) \neq 0$$

$$-2m^2 + 2m - 12 \neq 0$$

$$2m^2 - 2m + 12 \neq 0$$

$$(-1)^2 - 4 \cdot 1 \cdot 6 \neq 0 \quad m^2 - m + 6 \neq 0$$

$$\frac{1 + 24}{25} \quad m = \frac{1 \pm 5}{2} \quad \begin{matrix} \frac{6}{2} = 3 \\ \frac{-4}{2} = -2 \end{matrix} \quad \therefore S = \{m \in \mathbb{R} / m \neq 3 \text{ e } m \neq -2\}$$

$$c) \begin{cases} x - y = 2 \\ x + my = -z \\ -x + y - z = 4 \end{cases} \rightarrow \begin{cases} x - y + 0z = 2 \\ x + my + z = 0 \\ -x + y - z = 4 \end{cases} \Rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 1 & m & 1 \\ -1 & 1 & -1 \end{pmatrix}$$

$$\text{Det } A = \begin{vmatrix} 1 & -1 & 0 \\ 1 & m & 1 \\ -1 & 1 & -1 \end{vmatrix} \neq 0 \Rightarrow -m + 1 + 0 + 0 - 1 - 1 \neq 0$$

$$-m - 1 \neq 0$$

$$-m \neq 1$$

$$m \neq -1$$

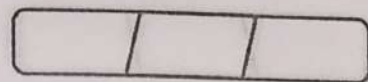
$$\therefore S = \{m \in \mathbb{R} / m \neq -1\}$$

$$d) \begin{cases} mx + y - z = 4 \\ x + my + z = 0 \\ x - y + 0z = 2 \end{cases} \rightarrow \begin{pmatrix} m & 1 & -1 \\ 1 & m & 1 \\ 1 & -1 & 0 \end{pmatrix} \Rightarrow \begin{matrix} -m - m + 0 + 0 - 1 - 1 \neq 0 \\ 2(-m) - 2 \neq 0 \\ -2m \neq 2 \end{matrix}$$

$$m \neq -2$$

$$\therefore S = \{m \in \mathbb{R} / m \neq -2\}$$





$$\begin{array}{l} 7 - \text{peça correta} = C = +6 \\ \text{peça com defeito} = D = -2 \end{array} \quad \left\{ \begin{array}{l} C + D = 225 \rightarrow 2C + 2D = 450 \\ 6C - 2D = 750 \quad 6C - 2D = 750 \\ \hline 8C + 0D = 1200 \end{array} \right.$$

*R: foram produzidas 150 peças corretamente

$$C = \frac{1200}{8} = 150$$

$\therefore C = 150$ peças

$$\begin{array}{l} 8 - C = \text{Distância carro} \rightarrow \\ B = \text{Distância Bicicleta} \end{array} \quad \left\{ \begin{array}{l} C + B = 540 \\ C \cdot 0,6 + B \cdot 0,2 = 300 \end{array} \right.$$

$\downarrow \div 2$

$$4500$$

$$-190$$

$$350$$

$$I - C = 540 - B$$

$$II - C \cdot 0,3 + B \cdot 0,1 = 150$$

$$540 - B \cdot 0,3 + 0,1B = 150$$

$$III - C = 540 - B$$

$$-0,3B + 0,1B = -390$$

$$C = 540 - 190$$

$$-0,2B = -390$$

$$\therefore C = 350 \text{ Km}$$

$$\therefore B = 195 \text{ Km}$$

*R: foram percorridos 350 Km de automóvel e 195 Km de bicicleta

$$\begin{array}{l} 9 - X = \text{cédulas de 5} \\ Y = \text{cédulas de 10} \\ Z = \text{cédulas de 2} \end{array} \quad \left\{ \begin{array}{l} X \cdot 5 + Y \cdot 10 + Z \cdot 2 = 500 \\ X + Y + Z = 92 \\ Y = Z \end{array} \right.$$

$$92$$

$$\times 5$$

$$460$$

$$I - \left\{ \begin{array}{l} 5X + 12Y = 500 \\ X + 2Y = 92 \end{array} \right. \xrightarrow{\times 5} \left\{ \begin{array}{l} 5X + 12Y = 500 \\ 5X + 10Y = 460 \end{array} \right.$$

$$0X + 2Y = 40$$

$$\therefore Y = 20 \text{ cédulas de 10}$$

$$II - \text{Se } Y = Z \\ 20 = Z$$

$$\therefore Z = 20 \text{ cédulas de 2}$$

$$III - X + Y + Z = 92$$

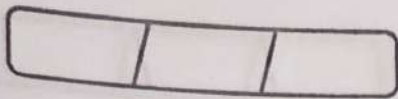
$$X + 20 + 20 = 92$$

$$X = 92 - 40$$

$$X = 52 \text{ cédulas de 5}$$

*R: Ele precisará de 52 cédulas de 5 para juntar a quantidade desejada





P

$$\begin{array}{l} 10 - K - Kiba \\ A - Akamuru \\ T - Tamaki \end{array} \quad \left\{ \begin{array}{l} K + A = 109 \\ K + T = 142 \\ T + A = 97 \end{array} \right. \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} K \\ A \\ T \end{pmatrix} = \begin{pmatrix} 109 \\ 142 \\ 97 \end{pmatrix}$$

$$\text{Det } P = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0 - 1 - 10 + 0 = -12$$

$$\text{Det } K = \begin{vmatrix} 109 & 1 & 0 \\ 142 & 0 & 1 \\ 97 & 1 & 1 \end{vmatrix} = 0 + 0 + 97 + 0 - 109 - 142 = -154$$

$$\therefore K = \frac{-154}{-12} = 12.83 \text{ kg}$$

$$\text{Det } A = \begin{vmatrix} 1 & 109 & 0 \\ 1 & 142 & 1 \\ 0 & 97 & 1 \end{vmatrix} = 142 + 00 - 97 - 109 = -64$$

$$\therefore A = \frac{-64}{-2} = 32 \text{ kg}$$

$$\text{Det } T = \begin{vmatrix} 1 & 1 & 109 \\ 1 & 0 & 142 \\ 0 & 1 & 97 \end{vmatrix} = 0 + 109 + 0 - 142 - 97 = -130$$

$$\therefore T = \frac{-130}{-2} = 65 \text{ kg}$$

